Solution Requirements

Date	23 October 2026
Team ID	0827CY231029
Project Name	Waypoint - AI Travel Itinerary Planner
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Functional requirements define the supported behaviors; non-functional requirements define operating targets and acceptance checks.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Email/password registration and login; issue a signed JWT; reject invalid or missing credentials.	High
2	Authorization levels	Authenticated users manage their own trips and preferences; public viewers only read a token-shared itinerary.	High
3	External interfaces	Use the configured Groq/OpenAI/Hugging Face/Ollama provider; connect to Supabase when configured; use SQLite locally.	High
4	Transaction processing	Create and persist a trip with days and activities; clone templates; update or delete saved data. Booking and payment transactions are outside scope.	High
5	Reporting	Show saved-trip dashboard and daily timeline; export JSON, Markdown and ICS.	High
6	Business rules	End date must not precede start date; budget must be non-negative; travelers range from 1 to 20; pace and travel style use allowed values.	High
7	Data privacy	Keep credentials and API keys on the server; store password hashes; scope private records to the signed-in user. Formal compliance certification is not established.	High
8	Other	Browse curated itineraries, save traveler preferences, rank destinations and request plan refinement.	High

Solution Requirements (continued)

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Keep routine API calls responsive; monitor AI latency separately.	Project target: routine API < 2 s; AI generation < 30 s. Not measured here.
2	Scalability	Deploy the API behind managed hosting and use a shared production database.	Horizontal scaling design; capacity requires load testing.
3	Security & Data Privacy	bcrypt passwords, signed JWTs, user-scoped queries and server-side provider secrets.	Unauthorized private-data requests rejected; use a strong production signing secret.
4	Reliability & Availability	Fall back to local itinerary rules if the chosen model fails or is absent.	Project uptime target: 99%+. No uptime evidence supplied.
5	Usability & Accessibility	Responsive React pages, readable timelines, clear inputs and loading/error states.	Review desktop/mobile layouts and keyboard use; formal accessibility audit pending.
6	Maintainability & Portability	Separate routers, services, repository adapters and React components.	Run locally with SQLite and deploy with Supabase using documented environment settings.

Scope and Acceptance Notes

Prices, transfers and activity suggestions are estimates. Verify current local availability before travel. Multi-city trips, simultaneous editing and a native mobile app are future work.

Project sources: 2. Requirement Analysis/README.md; backend schemas.py, security.py, main.py and services/ai_engine.py.